

$L = 20\ \mu\text{m}$ (channel), overlap $2\ \mu\text{m}$ each side (assumed); $W = 290\ \mu\text{m}$ (simulated width $1\ \mu\text{m}$)

Pd source, $70\ \text{nm}$

Pd drain, $70\ \text{nm}$

air (no passivation) [region 7]

IWO channel, $t = 2.0 / 6.3 / 13.2\ \text{nm}$ [regions 1 + 2 (0.25 nm interface layer)]

Al_2O_3 , $2\ \text{nm}$, $\epsilon_r = 9.0$ [region 3]

HfO_2 , $15\ \text{nm}$, $\epsilon_r = 19.57$ [region 4]

TiN gate (Conductor), $50\ \text{nm}$ [region 6, electrode "gate"]

not to scale; bottom-gate, top-contact TFT; $V_D = 0.7\ \text{V}$